As of: March 21, 2025
Received: March 14, 2025
Status: [redacted]
Tracking No. m89-c3t0-d6o2
Comments Due: March 15, 2025
Submission Type: API

PUBLIC SUBMISSION

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-1508
Comment on FR Doc # 2025-02305

Submitter Information

Email: [redacted]
Organization: Coalition for Content Provenance and Authenticity

---

General Comment

Good day. Thank you for the opportunity to respond to the Office of Science and Technology Policy's (OSTP) request for input on the Administration's development of an Artificial Intelligence Action Plan. On behalf of the Coalition for Content Provenance and Authenticity (C2PA), please find the attached submission. Should you have any questions, please be in touch at the contact information listed below. Thank you.

Attachments

C2PA AI Action Plan RFI Response_Final

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

C2PA

548 Market St
PMB 57274
San Francisco, California 94104

Friday, March 14, 2025

VIA ELECTRONIC FILING

AI Action Plan
Attn: Faisal D'Souza, NCO
415 Eisenhower Avenue
Alexandria, VA 22314

Dear Mr. D'Souza,

The Coalition for Content Provenance and Authenticity (C2PA) appreciates the opportunity to respond to the Office of Science and Technology Policy's (OSTP) request for input on the Administration's development of an Artificial Intelligence (AI) Action Plan ("Plan").

With the digital transformation of information sharing, the ability to establish the provenance (or source and history) of images, audio, video, and documents is critical. Today's capabilities—powered by AI and machine learning—have dramatically increased the ease, speed, and quality of alterations. Bad actors may exploit manipulated or entirely fabricated media to carry out cyber threats, commit crimes, exfiltrate sensitive economic data and intellectual property or undermine our national security. The Coalition for Content Provenance and Authenticity (C2PA), an open standards organization created by members of the Content Authenticity Initiative (CAI), Project Origin, and others through the Joint Development Foundation under the Linux Foundation, is collectively building an open technical standard to provide provenance and attribution for all forms of digital media to combat these, and many other, types of challenges.

The C2PA open technical standard can serve as a critical pillar in the Administration's efforts to enhance America's global AI dominance, promote human flourishing, economic competitiveness, and strengthen our national security. To that end, we offer the following background on the C2PA and Content Credentials, suggest possible use cases for the standard that are currently unexplored, including in the government, and provide several concrete actions the Administration should consider. Lastly, we provide examples of how private industry is leveraging this technology.

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

The C2PA – Building an Industry-led Global Standard

The C2PA is an industry-led, cross-sector coalition of technology companies and media organizations that are developing an open technical standard for disclosing and verifying the origin and/or history of digital content. Established in 2021, the C2PA has over two hundred members, and is guided by a steering committee consisting of Adobe, Amazon, BBC, Google, Intel, Meta, Microsoft, OpenAI, Publicis Groupe, Sony and Truepic. The C2PA open specification outlines how to establish digital content provenance using a technology called Content Credentials. Covering both technical details and normative guidelines, the standard aims to facilitate worldwide, opt-in adoption of digital provenance methods by cultivating a robust ecosystem of applications that support digital provenance for a diverse range of users and organizations while ensuring necessary security measures are met. Because the C2PA standard is openly available, anyone—including governments and their service providers—can use this standard to incorporate digital provenance information into their products and processes.

The C2PA standard is regularly updated based on new technologies, feedback from the community, and its needs, and is currently on version 2.1. The C2PA is currently fast-tracking its standard to become an International Standards Organization (ISO) standard. The C2PA expects ISO to approve the standard for publication by mid 2025.

Further, the C2PA's architecture is built to allow transparency across internet scale. Several unique tenets undergird the C2PA specification that make scale possible. We would like to highlight four critical aspects:

- **Interoperability**: The specification is written so that it can be adopted by any platform, OEM, or device. Any compliant mechanism can read, display, and add information to the provenance chain. This creates an interoperable provenance model for digital content that can move around the internet, with the content, in real-time.
- **Tamper Evidence**: Files signed with the C2PA specification become tamper-evident, helping content consumers know if unauthorized changes were made after signing occurred.
- **Multiple File Formats**: The specification can be applied to dozens of file formats ranging from image formats such as JPEG and PNG, to audio and video formats such as MP3 and MP4, as well as documents and text files. Also, provenance can be applied to machine learning files for data, models, and applications.
- **Voluntary**: For files captured with cameras, the addition of Content Credentials (the cryptographically bound metadata about the file's origin and history) is optional and can be added as creators decide. However, generative AI platforms are increasingly using

Content Credentials as a default setting on all generative outputs to disclose that their product/model created a given piece of media.

Other critical features such as a provenance chain for edits, the ability to adapt to newer generative technologies, stability, and scalability are the reasons why the C2PA open standard has grown in popularity with private industry and is soon to be a global ISO standard.

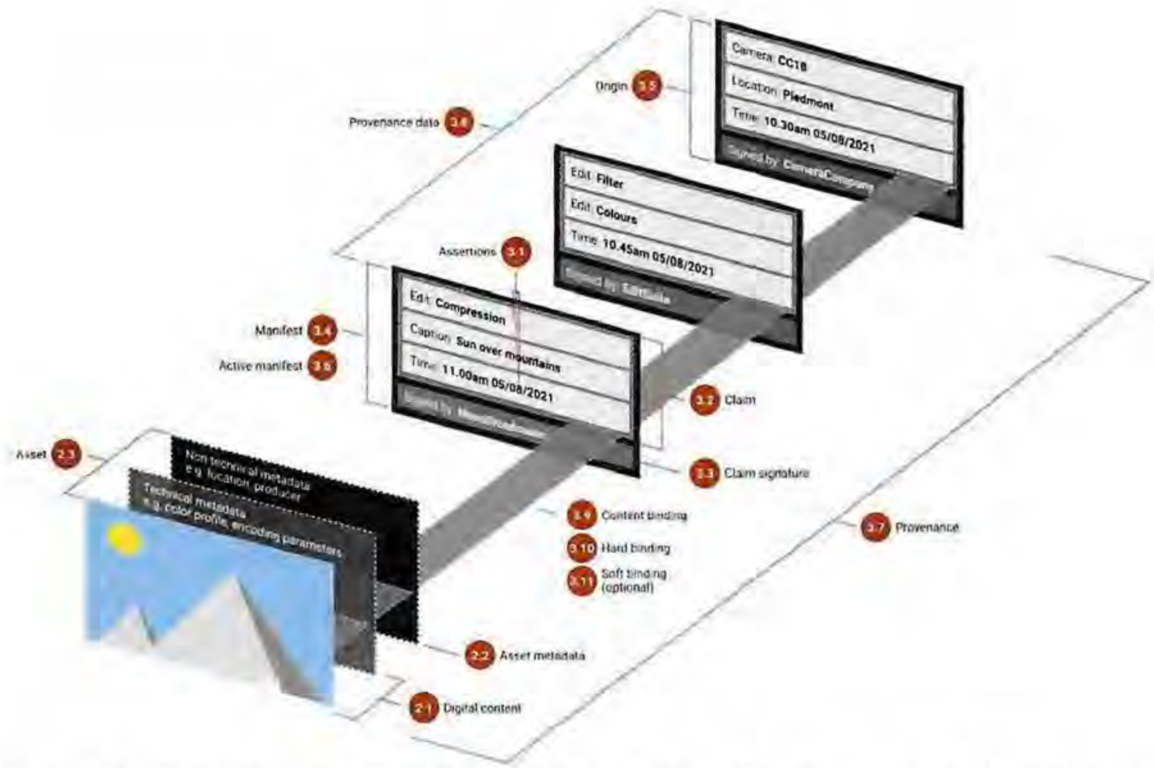

Figure 1 – This image shows how all these various elements come together to represent the C2PA architecture.

<u>Content Credentials</u>

The C2PA's Content Credentials are cryptographically secured metadata that enable individual creators, organizations, or even governments to embed information about themselves directly into media. They are designed to ensure the provenance and authenticity of digital content through verifiable claims using the same cryptographic technology that secures the Web, a crucial distinction for maintaining trust and transparency in the digital ecosystem. Additionally, Content Credentials can be incorporated either during media creation on hardware, like on phones or cameras, or when exporting through software. These credentials attach essential

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

metadata to a media file, allowing for the tracking of its origins, any subsequent modifications, and the tools or processes used in its production or alteration. Designed for global accessibility, Content Credentials include offline features—such as copying certificates to a secure enclave. The C2PA technical specification outlines the structure of these credentials and details the assertions regarding a media item's provenance.

To strengthen Content Credentials against tampering or removing their metadata, a digital watermark and fingerprint can be embedded into the digital media. This approach adds extra layers of preservation and recovery, resulting in what are known as Durable Content Credentials. Combined into a single approach, embedded metadata, watermarking, and fingerprinting form a unified solution that is robust enough to ensure that reliable provenance information is available no matter where a piece of content goes. This single, harmonized approach is the essence of Durable Content Credentials, and because it is the most comprehensive means of enabling transparency in content, we are seeing companies accelerate the adoption and integration of Content Credentials into their products, some which we detail below.

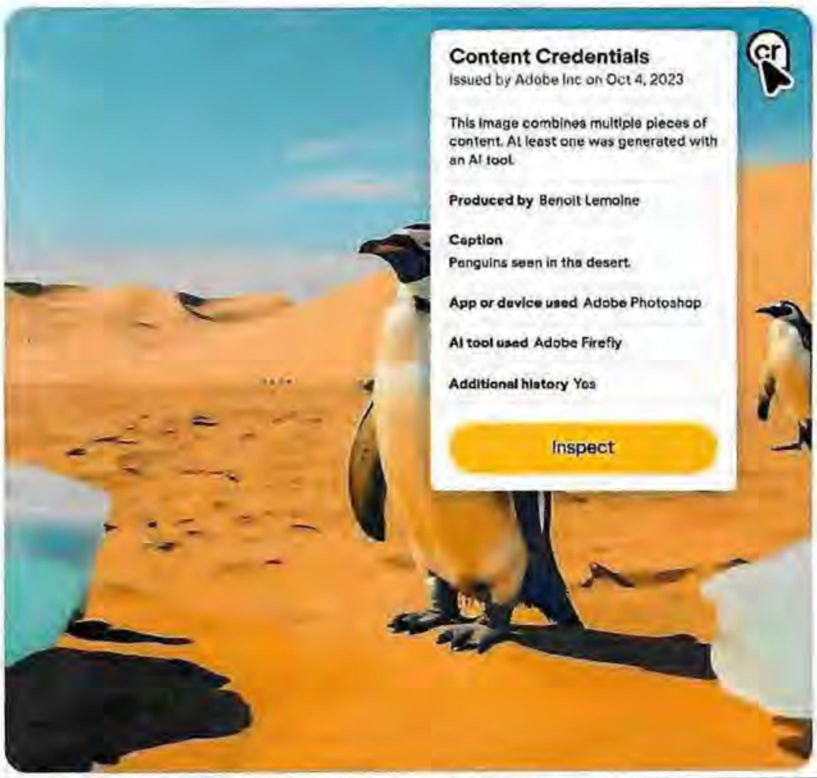

Fig. 2 – This image shows how embedded Content Credentials appear to users.

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

The C2PA & Content Credentials – Use Cases

The rapid reliance on digital content, combined with the explosive growth of synthetic media, has created new vulnerabilities, opening the door to fraud, image deception, and inaccurate decision making. This not only impacts consumers who depend on private enterprises for reliable products and services but also threatens American economic competitiveness by undermining the integrity of business operations in a global market.

Ultimately, private industry must adopt holistic and comprehensive cybersecurity and operational strategies to safeguard both their consumers and operations from these evolving threats. Emerging strategies for verifying digital content authenticity remain largely underexplored and unfamiliar to many enterprises, critical industries and governments.

Emerging technologies like image authentication, Content Credentials, and synthetic media detection are essential tools that should be actively explored and integrated (individually and in unison) by both the private sector and government as critical components of a robust defense framework. Below, we offer several use-cases for Content Credentials across the private sector and government.

Strengthen American innovation and economic growth: The United States can position itself at the forefront of artificial intelligence and drive innovation and economic growth by championing the use of open technical standards like the one from the C2PA. The C2PA not only fuels innovation and enhances competitiveness but it also provides a robust framework for safeguarding our trade secrets. Ensuring that digital assets carry content provenance metadata, like Content Credentials, can help protect them from theft and unauthorized replication. Moreover, the use of the C2PA standard to authenticate business and government information becomes a critical shield against adversaries, who might otherwise exploit or steal sensitive research and development data, thereby preserving the integrity of corporate and government innovations. This also extends to defending business and organizational reputations, as the C2PA framework helps prevent the misuse of synthetic media to fabricate damaging narratives about businesses, executives, government officials or entire industries. Globally, the establishment of a U.S.-led Content Credentials-based verification system can facilitate secure e-commerce and strengthen digital trade channels, and reinforce economic alliances and trade agreements, ultimately reducing the economic risks posed by adversarial states.

Enhance business security and mitigate costly fraud: The increased reliance on digital operations by businesses and enterprises today elevates their risk of fraud, unauthorized use, intellectual property theft, data manipulation, cyber-attacks, and other threats if they cannot ensure the accuracy and provenance of their data. Provenance and Content Credentials work hand-in-hand with cybersecurity frameworks to ensure that digital content is not tampered with. Companies dealing with sensitive data, such as financial institutions, healthcare providers, or law firms, can maintain clear digital audit trails for compliance and legal validation. Additionally, Content Credentials help mitigate fraud, including financial sector fraud, by verifying the authenticity of digital assets and transactions, reducing the risk of manipulated financial documents or unauthorized modifications. They can also serve as a critical defense against phishing campaigns by authenticating media embedded in phishing emails, enabling recipients to verify the legitimacy of images, videos, or documents before engaging with malicious content. With governments worldwide increasingly legislating new requirements for digital transparency, AI governance, and cybersecurity, the C2PA offers a proactive, industry-led solution to support compliance with these emerging regulations.

Protect national security: Adding provenance, transparency and image authenticity into business operations and processes will have positive impacts for innovation and national security. For example, the adoption of image authenticity and Content Credentials in CISA-designated critical industries—such as energy, telecommunications, financial services, and healthcare—strengthens American national security by protecting against cyber threats, deceptive media, and fraud. While Content Credentials can be used in crime scene photography and evidence collection, the same capability is also highly useful for national security and law enforcement forensics purposes. Moreover, Content Credentials can help establish the authenticity of scientific reports and images by helping prevent the manipulation of scientific data and research findings, which is crucial for national security-related research and development. Similarly, government and commercial imagery providers can implement Content Credentials for satellite imagery, providing a chain of provenance with their analysis, which is particularly important for intelligence applications that rely on accurate and verifiable imagery. Lastly, providing a means to verify the origin and editing history of digital content, Content Credentials can help combat the spread of foreign malign disinformation campaigns that may threaten national security. This is especially important in an era where AI-generated content is becoming increasingly sophisticated and prevalent.

Recently, the Department of Defense (DoD) started leveraging the C2PA standard and using Content Credentials for images on DVIDS, a public website for U.S. military digital media content. Having Content Credentials attached to its content helps DoD verify the authenticity of the images it captures and uses during important information gathering. In addition, the Content Credentials provide an important level of transparency in the content that the DoD shares externally — helping to build trust with the public.

Figure 3 – The Department of Defense's implementation of Content Credentials allows anyone to click the "CR icon" on images on DVIDS' website to get important context about the image including when and where it was captured.

Drive government and private sector efficiencies: Image provenance and transparency are critical economic drivers that can directly contribute to GDP growth and unlock incredible opportunities for innovation in the AI world. The C2PA's open, interoperable standard allows different industries and technology providers to integrate provenance into their ecosystems, helping businesses maintain credibility in their communications and digital assets, which in turn bolsters consumer confidence and increases economic opportunity.

Recommendations

We provide the following recommendations for your consideration:

Champion the use and adoption of C2PA's standard for content provenance and authenticity: The most beneficial approach and way to scale transparent digital content online is through the widespread adoption of Content Credentials by technology builders, content creators, and platforms. The OSTP and NIST's identification of and recognition of the C2PA's open standard will help increase adoption, scale the benefits of its adoption and enable the U.S. to shape the international standards debate around content provenance. To that end, the Administration should consider adoption of the C2PA standard for use in securing the government's digital media content.

Establish U.S. leadership by promoting an industry-led standard for the use of content provenance technologies: The Administration should promote a U.S. industry-led standard, like the C2PA, as the international standard on global content provenance frameworks. By focusing on driving adoption, best practices, information sharing amongst government agencies and our allies, the Administration can sustain and enhance America's AI dominance.

Conclusion

The C2PA technical standard, being industry-led, internationally recognized, and increasingly prevalent across companies and their products, uniquely positions the U.S. government to demonstrate global leadership by driving its adoption. Shaping standards is crucial for determining leadership in AI and other technological fields, and the U.S. government is uniquely positioned to lead by championing the work already undertaken by private industry in the C2PA, many of which are U.S.-headquartered companies.

In the world of content provenance and authenticity, the C2PA technical standard is one of the most compelling technical innovations to emerge. With American leadership, the C2PA can become a leading provenance standard worldwide.

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Appendix I

The C2PA & Content Credentials – Industry Leading the Way with Adoption

As generative AI continues to reshape the digital landscape, the need for transparency and trust in digital content has become increasingly critical. In response, major technology companies have been integrating Content Credentials into their platforms and products. From AI-generated images and videos to enterprise-level content management systems, these implementations represent a significant step toward enhancing digital content provenance. The following section highlights the adoption of Content Credentials by key industry leaders, showcasing their commitment to fostering greater transparency and authenticity in the evolving digital ecosystem.

Adobe: Adobe played a significant role in co-founding the C2PA, along with Truepic, Microsoft, Arm, Intel and the BBC, and since then, has been actively integrating Content Credentials into its products. For example, in Creative Cloud, consumers have access to Content Credentials in popular apps like Photoshop and Lightroom, where they can securely attach information such as names, dates, edits made and tools used. Other Adobe products including Illustrator, Express, Stock, Behance also support Content Credentials and Adobe is continuing to roll out capabilities across additional apps. For Adobe's generative AI model Firefly, Content Credentials are applied to select Firefly-powered features across Adobe apps to indicate the use of generative AI. Most recently, Adobe released a new free web app, Adobe Content Authenticity, designed to help creators protect and receive attribution for their work with Content Credentials.

Amazon: On September 12, 2024, Amazon announced it was joining the C2PA as a Steering Committee member. Amazon attaches Content Credentials to the visual assets produced using the company's generative AI models Nova Canvas and Titan Image Generator v1 and v2, which allows enterprise customers to create and edit images in a range of visual styles. Amazon is also working to incorporate Content Credentials in AWS Elemental MediaConvert, a file-based video processing service that transcodes content for broadcast and multi-screen delivery at scale. This implementation will allow its customers – such as news organizations and sports broadcasters – to communicate the provenance of the media they are sharing, allowing their distribution partners and news aggregators to verify the content's authenticity before posting it on their platforms.

Google: On February 8, 2024, Google, which also includes YouTube, joined the C2PA, marking a significant moment for bringing greater transparency to digital content. Google's C2PA membership is helping drive broader awareness of Content Credentials and furthering Content

Credentials adoption across the technology ecosystem, including in video. Last Fall, Google announced plans to expand their use of Content Credentials in Search, Ads, and YouTube – with more to come in 2025.

LinkedIn: On May 15, 2024, LinkedIn adopted the C2PA standard to enhance the transparency and authenticity of digital media on its platform. LinkedIn will begin displaying the "Cr" icon on images and videos that contain Content Credentials in the coming months. By clicking the icon, users will be able to view the origin of the media, including the source and history of the content, and whether it was created or edited by AI. LinkedIn is phasing its implementation on the platform and will soon include "Cr" display on ads.

Meta: On February 6, 2024, Meta announced it would ingest Content Credentials on Facebook, Instagram and Threads to assist with its initiative to increase transparency online by labeling AI-generated content. Meta's integration of Content Credentials helps users on Facebook, Instagram, and Threads to identify AI-generated or edited images. Meta joined the C2PA as a Steering Committee member in September of 2024.

Microsoft: Microsoft is actively integrating Content Credentials across its platforms to enhance transparency and trust in digital content. One of Microsoft's implementations of Content Credentials is within its Azure Text to Speech Avatar service. This feature attaches a manifest to generated avatar videos, indicating that the content was produced by Microsoft's service and providing a timestamp of creation. Microsoft has integrated Content Credentials into AI-generated images created with Bing Image Creator, Microsoft Designer, Azure OpenAI DALL-E, and Paint. Microsoft also has implementations to enable downstream display of Content Credentials. Microsoft released a browser extension that can validate C2PA assets. Our professional networking platform, LinkedIn, ingests provenance information based on the C2PA standard and displays associated provenance information for media in a user's feed.

OpenAI: OpenAI's joining of the C2PA steering committee on May 7, 2024, builds on OpenAI's existing C2PA integration for DALL-E 3, which was announced in February of 2024. Earlier in 2024, OpenAI began including Content Credentials in images generated and edited by DALL-E 3, the company's latest text-to-image model, in ChatGPT and the OpenAI API. In addition, the company also rolled out Sora, their new text-to-video generation model, in December 2024, announcing that Sora-generated videos come with C2PA metadata, which will identify a video as coming from Sora to provide transparency, and can be used to verify origin. OpenAI's membership and implementation of Content Credentials serves as a strong endorsement for the C2PA standard and advancing the collective mission to help restore trust in the digital ecosystem.

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Truepic: Truepic has taken a leading role in advancing the C2PA standard, becoming the first to support and integrate the C2PA specification into enterprise workflows through its Vision platform. As a founding member, Truepic offers enterprise-grade solutions that provide end-to-end content authentication and digital verifications for the world's leading businesses. By verifying data and embedding C2PA standards directly at the moment of capture, Truepic ensures authenticity immediately which allows enterprise to thrive in the AI era. Truepic's implementation of image and data verification aligning to the C2PA standard highlight the operational efficiency and innovations for business and government through authenticity and transparency.